Brain atlases and data structures :: Cheatsheet

Install, inspect and reshape ggsegverse atlases

Getting atlases

The ecosystem

Package	Holds
ggseg.formats	the atlas class, verbs and FreeSurfer I/O
ggseg	2D plotting (`geom_brain()`)
ggseg3d	3D plotting (htmlwidget)
ggseg.meshes	extra cortical and cerebellar surfaces
ggseg.extra	building new atlases, needs FreeSurfer
ggsegverse	installs and loads the lot

`dk()`, `aseg()`, `suit()` and `tracula()` ship inside ggseg and ggseg3d. Everything else lives in its own package on the r-universe.

Installing

```
options(repos = c(
  ggsegverse = "https://ggsegverse.r-universe.dev",
  CRAN = "https://cloud.r-project.org"
))

install.packages("ggsegverse")
```

```
library(ggsegverse)

ggseg_atlas_repos()          # everything available
ggseg_atlas_repos("schaefer") # matching a pattern
installed_ggseg_atlases()    # what you already have

install_ggseg_atlas("ggsegYeo2011")
install_ggseg_atlas_all()    # all of them, a large download
```

```
ggsegverse_packages()  # core packages
ggsegverse_deps()      # versions, and what is behind
ggsegverse_update()    # update the ones that are
ggsegverse_conflicts() # masked function names
ggsegverse_sitrep()    # one report covering all of it
```

Atlas packages follow the same shape: `library(ggsegYeo2011)` then `yeo7()`, and every atlas is a **function**.

The ggseg_atlas object

Anatomy

```
library(ggseg)
library(ggseg.formats)

names(dk())
#> atlas, type, palette, core, data
class(dk())
#> cortical_atlas, ggseg_atlas, list
```

Slot	Holds
`atlas`	the atlas name
`type`	`cortical`, `subcortical`, `cerebellar` or `tract`
`core`	one row per region: `hemi`, `region`, `label`
`data`	the geometry, keyed on `label`
`palette`	named vector of colours, one per label

`core` carries the metadata, `data` the drawing instructions, and the two are cross-checked on construction. Extra `core` columns (`lobe`, `names`) ride along and are available to `aes()`.

What goes in `data`

Type	Constructor	Contains
cortical	`ggseg_data_cortical()`	polygons + vertex indices
subcortical	`ggseg_data_subcortical()`	polygons + one mesh per region
cerebellar	`ggseg_data_cortical()`	flatmap polygons + SUIT vertices
tract	`ggseg_data_tract()`	centrelines + tangents

Vertex indices are **0-based** and point into the shared brain mesh.

Inspecting

```
atlas_regions(dk())
atlas_labels(dk())
atlas_views(dk())
atlas_type(dk())
atlas_geom(dk())
atlas_vertices(dk())
atlas_palette(dk())
atlas_geometry_type(dk())
```

```
is_ggseg_atlas(dk())
is_cortical_atlas(dk())
is_subcortical_atlas(aseg())
is_cerebellar_atlas(suit())
is_tract_atlas(tracula())
```

`plot(dk())` and `atlas_plot_palette(dk())` show the regions and the colours.

Reshaping atlases

Regions

Every verb takes an atlas and returns an atlas, so they pipe.

```
dk() |>
  atlas_region_keep("precentral") |>
  atlas_regions()
```

Verb	Does
`atlas_region_keep()`	keep matching regions
`atlas_region_remove()`	drop matching regions
`atlas_region_rename()`	rename, pattern to replacement
`atlas_region_contextual()`	mark as background context
`atlas_region_op()`	the general form behind these
`atlas_core_add()`	add columns to `core`

All take `(atlas, pattern)` and match on `region`; pass `match_on = "label"` to act per hemisphere.

Views

```
dk() |>
  atlas_view_keep(c("lateral", "medial")) |>
  atlas_views()
```

Verb	Does
`atlas_view_keep()`	keep these views
`atlas_view_remove()`	drop these views
`atlas_view_reorder()`	set the order
`atlas_view_select()`	keep one view per region
`atlas_view_gather()`	merge views into one
`atlas_view_remove_region()`	drop a region from one view
`atlas_view_remove_small()`	drop slivers below a threshold

`atlas_structure_reorder()` sets the order regions are drawn in.

CC BY-SA 4.0 · Athanasia Monika Mowinckel · ggseg.formats 0.0.4

Palettes

```
pal <- atlas_palette(dk())
head(pal, 3)

pal["lh_precentral"] <- "#5e3c58"
new_atlas <- set_atlas_palette(dk(), pal)
```

A palette must name **every** label in the atlas, so edit the existing vector rather than passing a short one. Set the type explicitly with `set_atlas_type(atlas, "cortical")` when an atlas you have built guesses wrong.

Data and meshes

Brain meshes

```
get_brain_mesh("lh", surface = "inflated")
get_cerebellar_mesh()

ggseg.meshes::get_cortical_mesh("lh", "pial")
ggseg.meshes::get_cerebellar_flatmap()
```

Each mesh is a list of `vertices` (`x`, `y`, `z`) and `faces` (`i`, `j`, `k`), at fsaverage5 resolution — 10,242 vertices per hemisphere. The inflated surface ships in **ggseg.formats**; the rest in **ggseg.meshes**.

Reading FreeSurfer output

```
read_freesurfer_stats("lh.aparc.stats")
read_freesurfer_table("aseg.stats")

read_atlas_files("path/to/atlas")
migrate_atlas_files("old/", "new/")
```

`read_freesurfer_stats(path, rename = TRUE)` tidies the column names into something that joins straight onto an atlas.

Converting and migrating

```
as_ggseg_atlas(x)      # anything -> ggseg_atlas
as_sf_atlas(dk())      # geometry as sf
as_polygon_atlas(x)    # geometry as polygons
convert_legacy_brain_atlas(old)
```

Renamed in 0.1–0.2. `brain_regions()`, `brain_labels()` and `brain_views()` became `atlas_regions()`, `atlas_labels()` and `atlas_views()`; `brain_atlas()` became `ggseg_atlas()`; `is_brain_atlas()` and `as_brain_atlas()` became `is_ggseg_atlas()` and `as_ggseg_atlas()`. The old names still work and warn.

Building a new atlas

ggseg.extra turns neuroimaging files into atlases. It needs FreeSurfer on the path.

```
library(ggseg.extra)

create_cortical_from_annotation("lh.aparc.annot")
create_cortical_from_gifti(...)
create_cortical_from_cifti(...)
create_subcortical_from_volume("aseg.mgz")
create_cerebellar_from_volume(...)
create_tract_from_tractography(...)
create_wholebrain_from_volume(...)
```

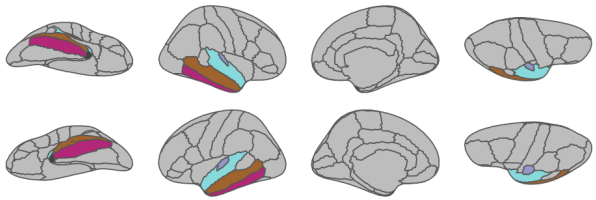
Then tidy the result with `atlas_smooth()`, `atlas_simplify()`, `atlas_dilate()` and `atlas_polish()`, and set up a package for it with `setup_atlas_repo()` and `use_atlas_github_actions()`. `sitrep()` reports whether the external tools it needs are present.

Checking your work

Does it render?

```
library(ggplot2)

dk() |>
  atlas_region_keep("temporal") |>
  brain_test_plot(
    position = position_brain(hemi ~ view)
  )
```



`brain_test_plot()` draws every region in its palette colour, so a region that has lost its geometry shows up as a hole. `plot(atlas)` is the quick version, and `atlas_plot_palette(atlas)` shows the colours on their own.

Validation

`ggseg_atlas()` checks on construction that:

- `type` is one of the four known types
- `core` has `hemi`, `region` and `label`
- every label in `data` appears in `core`, and the reverse
- the `data` container matches the declared `type`

A mismatch is a `cli` error naming the offending labels, not a silent coercion. `is_ggseg_atlas()` and the per-type predicates are the cheap checks to run first.

What an atlas package contains

An atlas package is a thin wrapper: the atlas object in `data/`, a function of the same name that returns it, and a `data-raw/` script recording how it was built. `setup_atlas_repo()` lays this out for you.

```
library(ggsegYeo2011)

yeo7()
atlas_regions(yeo7())
```

Gotchas

- Atlases are **functions**: `dk()`, not `dk`.
- Vertex indices are 0-based; R is 1-based. Add one before indexing a mesh yourself.
- A palette must cover every label, or `set_atlas_palette()` warns and the uncovered regions fall back to `na.value`.
- Cortical meshes share vertex indices across regions, so they cannot be decimated. Subcortical meshes can.
- `atlas_region_remove()` drops the region everywhere; use `atlas_view_remove_region()` to drop it from one view only.

Learn more

- `https://ggsegverse.github.io` — ecosystem documentation
- `https://ggsegverse.r-universe.dev` — every atlas package
- `https://github.com/ggsegverse` — source and issues

CC BY-SA 4.0 · Athanasia Monika Mowinckel · ggseg.formats 0.0.4